

A. Rainy day in Sylhet



Time Constraints

- **java** : 3.0 Seconds



Memory Constraints

- **java** : 256MB

Sylhet is one of the most beautiful cities in Bangladesh during the monsoon season, and Anik has decided to explore it. However, navigating the city is difficult due to poor visibility on cloudy days. He has K units of petrol, allowing him to travel one unit per fuel.

Given Sylhet's layout with interconnected cities and Anik's starting location, calculate and print the** possible** number of cities he can visit before running out of fuel.

Input

The input starts with four integers:

- C : Number of cities in Sylhet
- R : Number of roads connecting cities
- K : Fuel units remaining
- L : Starting city

Then R lines follow, each describing a bidirectional road between two cities (u, v).

Output

Output a single integer: total number of cities Anik can reach using up to K fuel units. Do not use `println`, use `print`.

Constraints

- $(2 \leq C \leq 10^3)$
- $(0 \leq R \leq \frac{C(C-1)}{2})$
- $(1 \leq K \leq 10^6)$
- $(1 \leq L \leq C)$

Sample Input

```
5 5 2 1
5 3
2 4
3 4
1 5
2 5
```

Sample Output

```
4
```

Explanation

- Anik starts at city 1.
- With 2 units of fuel, he can reach cities 5 and then 3 or 2.
- The total number of reachable cities is {1, 5, 3, 2}, hence the output is 4.